TANKSHIFT – MASTER PACK (REVISED)

India's water tanks are already a battery. We just switch it on.

Prepared for: Climate Tech Launchpad Program, Season 2 (MBRDI × T-Hub), REVA University, 2 November 2026

Team: [Name 1], [Name 2], [Name 3] – REVA University

Theme: Climate Intelligence & Energy Resilience

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Part 1 – How the Idea Was Chosen and Stress-Tested

16 ideas were scored by a simulated panel (incubation manager, MBRDI engineer, climate-policy expert, city official, VC). Finalists were checked online. Rejected with evidence: second-life battery cooling hubs (WRI, Jodhpur), e-waste metro kiosks (Cashify), rider cooling vests (Zomato, Swiggy), plastic roads, pothole apps, battery swapping, retrofits, flood alerts, heat insurance.

A second stress test against the Karnataka tariff order moved the target from residential to large HT consumers. Reason: residential pumps in Karnataka are on LT-1 with no ToD, while HT consumers ≥ 500 kVA have compulsory ToD with a ₹2/unit swing between night and peak.

Part 2 – The Idea in One Page

Every campus, hospital, tech park and factory pumps water into tanks and uses it hours later. That gap is energy storage nobody is using. Today pumps run on float switches whenever a tank drops, often in the 6–9 am and 6–10 pm windows where Karnataka charges ₹1 extra per unit, and never deliberately in the 10 pm–6 am window where it gives ₹1 back.

TankShift is a controller that plugs into the existing pump starter, fills the tanks in the cheap window, blocks the expensive windows, keeps a safety reserve, and reports savings on the actual ToD bill. The site saves energy charges and demand charges. The DISCOM gets load moved off its evening peak without buying a battery.

Stages:

- Stage 1 – campuses and institutions.
- Stage 2 – tech parks and industry, with a virtual-power-plant offer to BESCOM.
- Stage 3 – residential when domestic ToD arrives, and EV charger scheduling on the same box.

Part 3 – Pitch Deck (9 Slides)

Slide 1 – Title & Team

TankShift. India's water tanks are already a battery. We just switch it on. Theme: Climate Intelligence & Energy Resilience. Team, REVA University.

Slide 2 – Problem

Large campuses pump thousands of units a day on float switches, blind to the clock. Karnataka's compulsory ToD charges HT consumers ₹1 extra per unit from 6–9 am and 6–10 pm and refunds ₹1 from 10 pm–6 am, December to June. Demand charges of ₹350–380 per kVA punish pumps that start during the site's peak. Facility teams have no cheap tool to act on either. The grid pays too: evening peaks run on the dirtiest plants.

Slide 3 – Solution

A retrofit controller for the pump starter that fills tanks in the rebate window, locks out the surcharge windows, protects a reserve, and proves the saving on the bill. No lithium, no new civil work.

Slide 4 – Design & USP

Level sensors in sump and OHT; 3-phase energy meter on the pump feeder; controller holds the local ToD table and the site's demand profile; contactor driven through the existing Auto/Manual starter so the watchman's manual override still works. USP: timers don't know the tariff or the site peak; building-management systems cost lakhs and campuses this size don't have them; the firmware adapts to any state's ToD table.

Slide 5 – Visual

Karnataka clock bar: red 6–9 am and 6–10 pm (+₹1), green 10 pm–6 am (−₹1), grey day. Old pump runs scattered; TankShift runs in green. Site section: borewell/transfer pump, sump, OHT, controller on the starter panel, dashboard "Last month: 31,000 units moved, ₹44,000 saved, 18 kVA off peak".

Slide 6 – Market

Karnataka: every HT-2 consumer ≥ 500 kVA is a mandatory ToD customer: universities, hospitals, IT parks, factories, hotels, malls. India: 20+ states with ToD for HT; Maharashtra already pays solar-hour rebates. Water boards (HT-1) as enterprise accounts: BWSSB's power bill is ₹75 crore a month. Who pays: the facility (hardware + annual licence), later the DISCOM (demand-response fee).

Slide 7 – Competitors

Float switches and timers: blind to tariff and demand. BMS vendors (Schneider, Honeywell, Siemens): ₹10–50 lakh, for buildings that already have energy managers. Energy auditors: a report, not a device. Our line: we are the ₹10,000 version of a BMS feature, for the 500 kVA–5 MVA sites nobody serves.

Slide 8 – Grant

Step 1, prove it: 3 pump sets on REVA campus + 2 at a partner institution, 90 days, ₹36,000. Step 2, only if Step 1 shows savings on the ToD bill: 2 more sites + dashboard, ₹14,000. Unused money returned.

Slide 9 – Conclusion

The grid's evening peak is India's most expensive hour. Campuses have thousands of units of flexible pumping and don't know it. TankShift finds it, moves it, and proves it on the bill. Ask: ₹36,000 to run it on REVA's own meter.

Speaker Notes – Judge Questions

- **"Isn't this just a timer?"** A timer doesn't know the ToD table, the season switch in July, the tank levels or the site's demand peak.

- **"Big sites already do this."** Some with a BMS do. Most universities and hospitals run pumps on floats. We measured it on our own campus.
- **"What about residential?"** Not yet. Karnataka residential pumps have no ToD. When the central rule lands in 2028, same box, new tariff table.
- **"Climate link?"** Every unit moved off the 6–10 pm peak is a unit not generated by a peaking plant.
- **Don't know?** "That's what the 90-day pilot measures."

Part 4 – Report (Key Sections)

4.1 Problem

As Slide 2. Add: the July–November zero table means the device must also deliver demand-charge savings and grid-friendliness in monsoon, not only ToD arbitrage.

4.2 How It Works

Logic priority:

1. Reserve level – pump now.
2. Pump in the rebate window.
3. Never pump in surcharge windows.
4. Spread night pumping to avoid creating a new night demand peak.
5. Learn the site's demand curve and keep pump starts off its monthly peak.
6. Accept DISCOM hold signals.

4.3 Why It's New

Load shifting is known in large BMS installations. No low-cost, pump-specific, tariff-table-driven retrofit exists for mid-size HT consumers in India.

4.4 Market and Economics

One campus: 1,000–1,500 units/day; ~₹1.4/unit swing for 7 months ≈ ₹3–4 lakh/yr energy; ₹1–2 lakh/yr demand charges. Hardware ₹8–12k installed. Payback in weeks. Tech park with 5,000 units/day: ₹15 lakh+/yr. All figures are estimates to be replaced by measured pilot data.

4.5 Money

Step 1 – Prove it (₹36,000)

Item	₹
5 HT-grade kits at ~₹5,500 parts	27,500
Electrician + panel work	4,000
Cloud and SIM data	2,000
Contingency	2,500
Total Step 1	36,000

Step 2 – Scale it (₹14,000, only if Step 1 shows savings on the ToD bill): 2 more sites + dashboard. **Total both steps: ₹50,000.**

4.6 Risks

Risk	Answer
Night pumping creates a new demand peak	Controller staggers pump starts and caps simultaneous pumps
Monsoon months have zero ToD	Demand-charge logic still works; grid-friendly scheduling continues
Site already has a BMS	Not our customer; we target the 500 kVA–5 MVA sites without one
Tank runs dry	Hard reserve; worst case it behaves like the float switch
Tariff tables change	Table is a downloadable file, not firmware
Fire tanks used as buffer	Never – fire tanks must stay full by law and are excluded

4.7 Impact

- **Climate:** units moved off the evening peak.
- **Economic:** payback in weeks, no capex for the DISCOM.
- **Social:** hospitals and universities first.
- **Circular:** zero new storage material.

Part 5 – Master Plan

Stage	What	Status
1	Idea, judge testing, novelty check, tariff stress test	Done
2	9-slide deck	Done
3	Report	Done
4	Working plan	Done
5	Circuit, parts list, safety checks (HT version)	Done
6	2D blueprint, 3D enclosure	Done
7	Firmware logic	Pending
8	Application-form answers	Pending
9	REVA bill analysis	Pending
10	Tabletop demo for Nov 2	Optional

Before Nov 2:

- Day 1: team names; rehearse the 3-minute pitch twice.
- Day 2: read the T-Hub Season 2 page and application form; list missing answers. Get REVA's last 6 electricity bills from the estate office and read the ToD slot consumption – that single page is the Slide 2 evidence.
- Day 3: print circuit and blueprint as backup pages.
- Day 4 (optional): tabletop demo – ESP32 + float switch + relay + LED that turns on only inside a set "night window".
- Day 5: rest.

90-day pilot: Weeks 1–3 parts and assembly, bench-test every kit on a lamp. Week 4 installation with electrician. Weeks 5–16 logging, weekly savings check, collect bills. Week 17 report and BESCOM meeting.

Roles (team of 3): hardware + electrician; firmware + app; sites, data + pitch.

Part 6 – Circuit Design (HT Site Version, Verified)

Design rules as before:

- No ESP32 pin sees more than 3.3 V.
- No boot-strapping pin (GPIO 0, 2, 12, 15) is used.
- The relay never switches the motor; it only drives the starter/contactor coil circuit.
- Mains and logic separated by an isolation barrier.
- Manual bypass keeps the pump running if the controller dies.
- Dry-run protection in hardware (sump float) and software.
- RTC keeps time without internet.

Parts list (3-phase campus pumps)

Part	Spec / role	₹
ESP32 DevKit V1	Brain	450
HLK-10M05	230 V → 5 V 2 A	350
DS3231	RTC	120
3-phase energy meter, Modbus RTU (Eastron SDM630 / Selec EM2M class) + 3 CTs sized to pump FLC	Measures pump feeder	4,500–5,500
MAX485 module	Modbus to ESP32 (GPIO 16/17, DE/RE on GPIO 5)	60
Relay module, opto, 5 V, HIGH trigger	Drives starter coil circuit	60
Interface to existing DOL/star-delta starter	Relay contact in series with starter Auto position; no new contactor if starter exists	0
If no starter: 3-pole contactor sized to motor, 230 V coil + MPCB	Switches pump	1,500–3,000
Float switches ×4, JSN-SR04T, resistors, caps, LED, button	As before	900
IP65 enclosure, DIN rail, glands	Housing	450

Pin map: unchanged from the single-phase version except the meter – GPIO 16 RX / GPIO 17 TX to MAX485, GPIO 5 for DE/RE. PZEM-004T is dropped for 3-phase.

GPIO	Connected to	Note
16 / 17	MAX485 RO / DI	Modbus to energy meter
5	MAX485 DE/RE	Transmit enable
21 / 22	DS3231 SDA / SCL	I^2C
25 / 27	JSN-SR04T TRIG / ECHO (via 1k/2k divider)	ECHO is 5 V
26	Relay IN, 10k pull-down	Relay off at boot

GPIO	Connected to	Note
32 / 33	OHT FULL / RESERVE floats	Internal pull-up
34 / 35	Sump LOW / HIGH floats	External 10k pull-up
14	Status LED via 330 Ω	
4	Test button to GND	Internal pull-up

Mains wiring in words: the existing starter keeps its Auto/Manual switch. In Auto, the starter coil is fed through TankShift's relay contact in series with the existing float-switch contact (both must say yes). In Manual, the watchman bypasses both. Energy meter CTs go on the three phases of the pump feeder, voltage taps through 1 A fuses. Earth bonding unchanged. The electrician does the panel work; the controller's low-voltage half is yours.

Part 7 – Blueprint and Enclosure

Same drawings as before. Replace "pump 1.5 HP, 230 V" with "transfer/borewell pump, 3-phase, 7.5–30 HP" and place the controller on or beside the existing starter panel in the pump house. Fire tanks are not part of the buffer.

Enclosure: 300 × 200 × 130 mm, IP65. Back half (mains, DIN rail) and front half (3.3/5 V logic) split by a full-height 3 mm acrylic/polycarbonate barrier; only the meter's serial cable crosses. Glands on the bottom face for mains in, pump/starter link and sensor bundle; earth stud beside the mains gland. Label every wire at both ends.

Part 8 – Still To Do

- Firmware logic (state machine, settings, offline tariff table, watchdog).
- Application-form answers.
- REVA bill analysis.
- Optional tabletop demo (LED + relay inside a set night window).

Sources

KERC Tariff Order 27 March 2025, Electricity Tariff-2026 annexure (ToD tables, LT-1, HT-2, HT-4, demand charges); Ministry of Power ToD rules and 2026 explainers; Deccan Herald / The Week on BWSSB; WRI, CEEW, Cashify, Outlook Business, Business Standard for rejected ideas.